

WORKSHOP: FROM ZERO TO WEB-AUDIO: BUILDING CSOUND-POWERED APPS WITH DR.C

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ABSTRACT

This workshop provides a fast-paced, hands-on introduction to Dr.C, a specialized AI environment for developing and deploying Csound-based applications within the Web Audio API. While Csound offers unparalleled DSP depth, the barrier to entry for modern web deployment can be high. Dr.C removes these friction points, allowing developers to focus on synthesis and interface design rather than boilerplate integration. In this 2-hour session, participants will be guided through a speed-run of the Dr.C workflow, transitioning from executing Csound code directly in the browser to building interactive audio-reactive web applications. The workshop culminates in a mini-hackathon where attendees compete to build full-featured audio applications, followed by a live showcase and peer-voting for the best application.

Keywords: Csound, AI-Collaboration, Web-Audio, Rapid-Prototyping, Human-in-the-Loop

1. WORKSHOP OVERVIEW

Dr.C is an open-source, domain-specialized AI assistant for Csound development that implements an alternatives-first design paradigm, multi-tier knowledge retrieval, and semantic validation framework [1]. This workshop demonstrates how Dr.C accelerates Csound-based web application development from concept to deployment.

1.1. Learning Objectives

By the end of this workshop, participants will be able to:

1. Execute Csound code directly in web browsers using Dr.C's export workflow
2. Map real-time data sources (user input, sensors, APIs) to Csound synthesis parameters
3. Design and customize audio-reactive user interfaces for web deployment
4. Apply Dr.C's alternatives-first workflow for rapid synthesis exploration
5. Deploy standalone Csound-powered web applications

1.2. Target Audience

This workshop is designed for:

- Csound users interested in web deployment workflows
- Web developers curious about integrating advanced audio synthesis
- Music technologists exploring AI-assisted development tools
- Educators seeking new pedagogical approaches for computer music

Basic familiarity with Csound syntax is helpful but not required. Web development experience (HTML/JavaScript) is beneficial but not mandatory.

2. WORKSHOP STRUCTURE

2.1. Introduction (15 minutes)

Topics covered:

- Dr.C architecture overview: multi-agent system, specialized tools, knowledge retrieval
- Csound-Web integration via @csound/browser CDN
- Alternatives-first design paradigm demonstration
- Quick Start workflow walkthrough

The introduction showcases Dr.C's core capabilities through live demonstration of the alternatives workflow. Participants observe how Dr.C proposes multiple synthesis approaches (technique, character, complexity) before execution, maintaining human creative control throughout the design process.

2.2. Guided Code-Along (30 minutes)

Hands-on activities:

1. **Basic synthesis instrument (10 min):** Build a simple interactive synthesizer using Dr.C's alternatives workflow. Explore FM synthesis variations with different modulation ratios and envelope designs.
2. **UI element connection (10 min):** Map sliders, knobs, and buttons to Csound parameters. Learn about Cabbage widget bindings, channel declarations, and parameter smoothing.
3. **Audio-reactive visualization (10 min):** Connect synthesis parameters to visual elements. Implement real-time amplitude tracking for particle systems or waveform displays.

All code is generated collaboratively with Dr.C, demonstrating how the AI proposes alternatives while participants maintain creative decision-making authority.

2.3. The Dr.C Hackathon Challenge (60 minutes)

Challenge brief: Design and deploy an audio-reactive web application using Dr.C.

Suggested project categories:

- Generative music systems with autonomous instrumental layers
- Data sonification applications (weather, stocks, sensors)
- Interactive synthesis explorers with parameter visualization
- Algorithmic composition tools (L-systems, cellular automata)
- Audio effects processors with visual feedback

Participants work independently or in small teams, using Dr.C to rapidly prototype, iterate, and refine their applications. Instructors circulate to provide guidance, answer questions, and demonstrate advanced Dr.C features.

Judging criteria:

- Creative use of synthesis techniques
- Quality of audio-visual integration
- Interface design and usability
- Technical ambition and execution

2.4. Showcase & Awards (15 minutes)

Lightning presentations: Each participant/team presents their application in 60-90 seconds, demonstrating key features and sonic characteristics.

Peer voting: Attendees vote for their favorite applications across categories: Most Creative, Best Sonic Design, Best UI/UX, and Best Overall.

Reflection: Brief discussion of design decisions, challenges encountered, and insights gained from the alternatives-first workflow.

3. TECHNICAL REQUIREMENTS

3.1. Instructor Equipment

- Laptop with HDMI output for projection
- Stereo sound system for high-quality audio playback
- Reliable internet connection for live demonstration

3.2. Participant Requirements

- Personal laptop (macOS, Linux, or Windows)
- Headphones for individual audio monitoring during hackathon
- High-speed internet access (essential for Dr.C browser-based workflow)
- Modern web browser (Chrome, Firefox, Safari, or Edge)

Pre-workshop preparation: Participants will receive instructions for accessing Dr.C's web interface and optional pre-reading materials on Csound basics and Web Audio API fundamentals.

3.3. Room Setup

- Tables or desks with power outlets for laptops
- Sufficient space for comfortable laptop operation and note-taking
- Projector screen visible from all seating positions
- Audio system positioned for even sound distribution

4. PEDAGOGICAL APPROACH

This workshop demonstrates Dr.C's "vibe coding" pedagogy, where participants describe aesthetic goals ("warm, resonant filter," "bright percussive attack") rather than technical implementation details. Dr.C translates these descriptions into Csound code while providing educational scaffolding through:

- **Alternatives with explanations:** Each proposed design includes technical rationale and parameter semantics
- **Signal flow diagrams:** ASCII visualizations of processing chains demystify code structure
- **Design tree exploration:** Non-linear version control captures all explored alternatives for later review
- **Immediate sonic feedback:** Instant audio rendering enables learn-by-listening workflows

The alternatives-first approach ensures participants remain active decision-makers throughout the creative process, learning synthesis concepts through guided exploration rather than syntax memorization.

5. EXPECTED OUTCOMES

Based on educational deployment at Berklee College of Music across Composition (EP461), DSP (EP413), and Csound (EP337) courses, we anticipate the following workshop outcomes:

1. **Immediate productivity:** Participants create working applications within the 2-hour timeframe, demonstrating Dr.C's ability to compress development cycles from weeks to hours.
2. **Conceptual understanding:** Post-workshop surveys reveal increased comprehension of synthesis techniques, signal flow, and parameter relationships through hands-on experimentation.
3. **Motivation for deeper learning:** The ability to rapidly realize creative ideas motivates participants to explore Csound documentation, opcode libraries, and synthesis theory more deeply.
4. **Portable skills:** Experience with Dr.C's alternatives-first workflow transfers to other creative technical domains beyond Csound.

6. RELATIONSHIP TO COMPANION PAPER

This workshop accompanies the LAC 2026 paper "Dr.C: A Human-in-the-Loop Agentic AI for Csound," which presents the theoretical foundations, system architecture, and educational research supporting Dr.C's design. The workshop provides hands-on demonstration of concepts discussed in the paper:

- **Alternatives-first paradigm** (Paper Section 5): Live demonstration of design space exploration and creative agency preservation
- **Semantic validation framework** (Paper Section 4): Participants experience automatic error detection for Cabbage contracts and Csound 7 correctness
- **Multi-tier knowledge retrieval** (Paper Section 3.3): Access to opcode cards, CSD examples, and synthesis textbook knowledge during development
- **Educational deployment outcomes** (Paper Section 8): Workshop replicates pedagogical approaches proven effective in university courses

The workshop translates Dr.C's research contributions into practical, immediately applicable skills for LAC attendees.

7. ABOUT THE INSTRUCTOR

Dr. Richard Boulanger is Professor of Electronic Production and Design at Berklee College of Music and editor of *The Csound Book: Perspectives in Software Synthesis, Sound Design, Signal Processing, and Programming* (MIT Press, 2000) [1]. He has taught Csound, DSP, and computer music composition for over 30 years and co-developed Dr.C with Mateo Larrea (CCRMA, Stanford University) to address fundamental limitations in general-purpose AI coding assistants for domain-specialized creative work.

8. CONCLUSION

This workshop offers LAC attendees practical experience with a novel approach to AI-assisted Csound development. By combining hands-on coding, competitive hackathon challenges, and peer showcase presentations, participants gain transferable skills in web audio deployment while exploring the creative possibilities of alternatives-first AI collaboration.

Dr.C is freely available under MIT license at <https://github.com/csounder>, ensuring ongoing access to tools and resources introduced during the workshop.

9. ACKNOWLEDGMENTS

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10. REFERENCES

- [1] R. Boulanger, Ed., *The Csound Book: Perspectives in Software Synthesis, Sound Design, Signal Processing, and Programming*. MIT Press, 2000.
- [2] V. Lazzarini, S. Yi, J. Heintz, Ø. Brandtsegg, and I. McCurdy, *Csound: A Sound and Music Computing System*. Springer, 2016.
- [3] "Csound – A Sound and Music Computing System," <https://csound.com/>, accessed March 2026.
- [4] "The Csound Reference Manual," <https://csound.com/docs/manual/index.html>, accessed March 2026.
- [5] R. Walsh, "Cabbage Audio – Documentation and Examples," <https://cabbageaudio.com/docs/>, accessed March 2026.
- [6] J. Heintz, A. Hofmann, and I. McCurdy, "FLOSS Manual – Csound," <https://flossmanual.csound.com/>, accessed March 2026.